



SimToReal

Simulation to real deployment platform for RL based robots.

Application Scenario



Framework





Scientific problem, technical challenge

Mostly robots are trained inside simulator like mujoco. Once training done, trained policy needs to be moved into the real robot. Currently it's done by someone manually, so I'm trying to make it automatically convert it to the runnable format in real robot through a python library.

Challenges

- Different formats for different policies
- Sensors and Actuators read/write functions are different



Objectives

- **Phase 1**
 - Automatically convert the trained policy into hardware runnable format
- **Phase 2**
 - Automatically generate a code that connects the robots actuators and sensors with the trained policy
- **Phase 3**
 - Directly upload the generated files to the hardware via wireless transfer.

Currently in the 2nd phase of the project



Methods

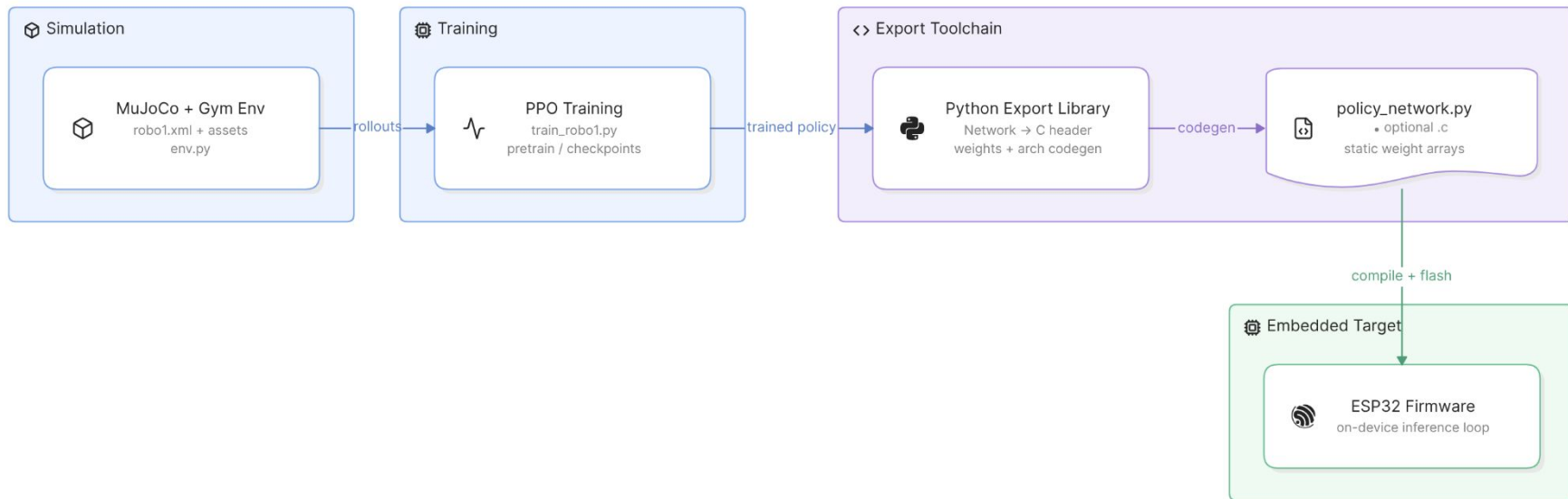
- Python library

Using a Python library streamlines the workflow by providing built-in functions to convert trained data. This eliminates the need to manually modify the code each time the robot is trained.

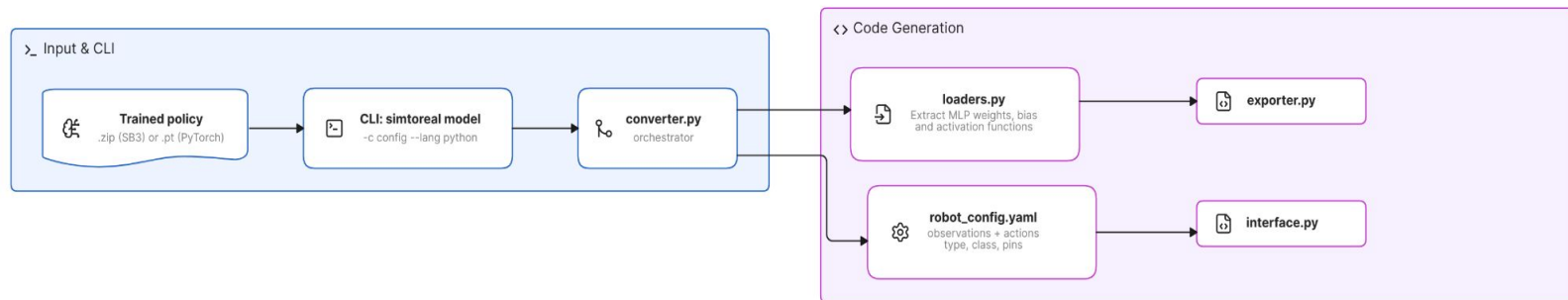
- BLE

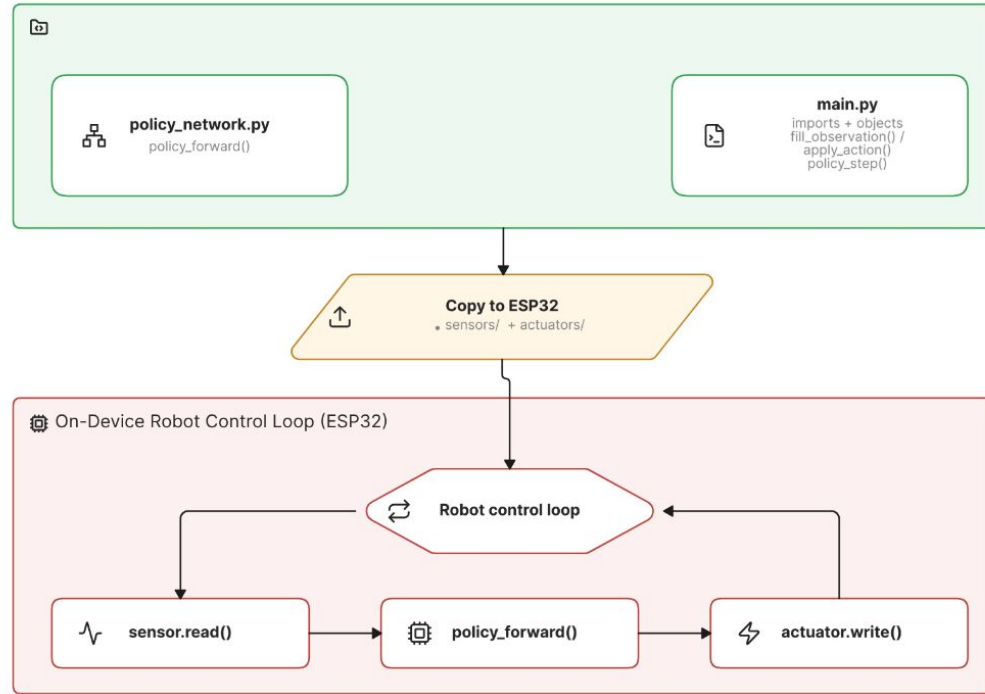
Uploading the files to the ESP32 via BLE easily

system design and implementation



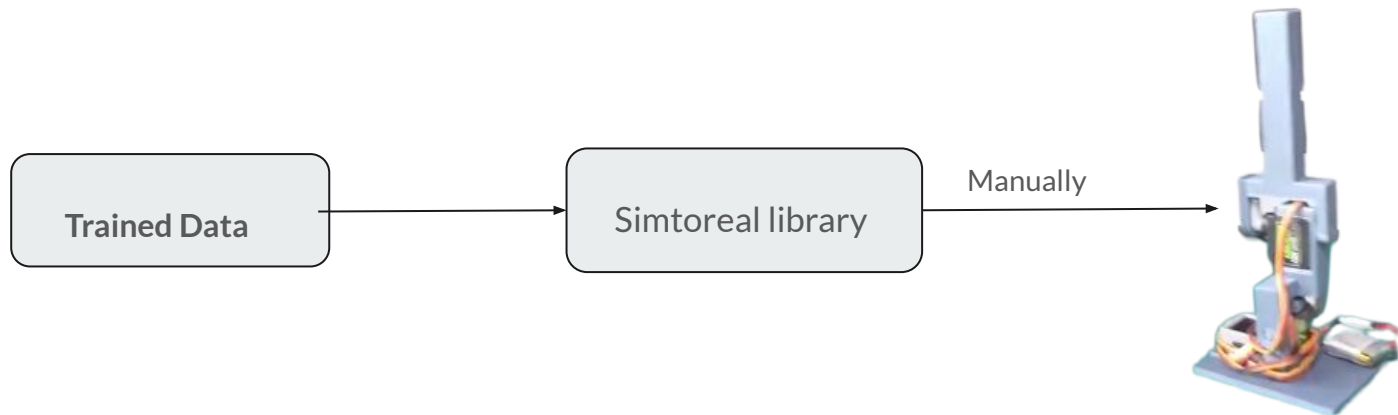
How the library works





Practical Demonstration

Using a tiny 2 DOF robot showing that code works



Thank you . . .